FUNCTIONAL ANALYSIS

(5 hours)

January 10th 2020

Please write on the front page your preferred time for the oral exam.

1. Let c be the space of convergent complex sequences with the norm

$$\|(x_n)\| = \sup_{n\geq 1} |x_n|, \quad (x_n) \in c.$$

a) Prove that c is a closed subspace of l^∞.
b) Let $l : c \to \mathbb{C}$ be the linear functional

$$l((x_n)) = \lim_{n\to\infty} x_n, \quad (x_n) \in c.$$

Show that $l \in c'$, but there is no sequence $(y_n) \in l^1$ such that

$$l((x_n)) = \sum_{n\geq 1} x_n y_n, \quad (x_n) \in c.$$

2. Let $\{t_1, \ldots, t_n, \ldots\}$ be a countable dense set in $[0,1]$. Define the linear map $T : C([0,1]) \to l^\infty$ by

$$Tf = (f(t_n)), \quad f \in C([0,1]).$$

a) Show that T is isometric and not surjective.
b) Show that there exists $l \in (l^\infty)'$ such that for every $f \in C([0,1])$ we have

$$\int_0^1 f(t)dt = l(Tf).$$

Hint: Define first such an l on the range of T.

3. Let X be a Banach space and let $T : X \to X$ be a linear map. Show that T is continuous if and only if for every sequence (x_n) in X that converges weakly to $x \in X$ we have that (Tx_n) converges weakly to Tx.

4. Consider the Banach space $C^2([0,1])$ of twice differentiable functions f on $[0,1]$ with the norm

$$\|f\|_2 = \sup_{t\in[0,1]} |f(t)| + \sup_{t\in[0,1]} |f'(t)| + \sup_{t\in[0,1]} |f''(t)|.$$

Let $a, b \in C([0,1])$ and assume that for every $g \in C([0,1])$, the differential equation

$$f'' + af' + bf = g$$

has a solution $f \in C^2([0,1])$. Prove that there exists $c > 0$, such that for every $g \in C([0,1])$, the above equation has a solution $f \in C^2([0,1])$ with $\|f\|_2 \leq c\|g\|$.
Hint: Use the equation to define a map $M \in \mathcal{L}(C^2([0,1]), C([0,1]))$ and analyze its properties.

5. Let $T : C([0,1]) \to C([0,1])$ be defined by
$$Tf(x) = \frac{1}{x^2}\int_0^x tf(t)dt,\ x \neq 0, \quad Tf(0) = \frac{f(0)}{2}.$$
a) Show that T is bounded but not compact. You do not have to prove that T is well defined.
Hint: compute as many eigenvales as you can.
b) Let $M : C([0,1]) \to C([0,1])$ be defined by $Mf(x) = xf(x)$. Prove that TM is compact.

6. Let H be a separable Hilbert space and let $C : H \to H$ be a compact symmetric linear map.
(i) Prove that the spectrum of C is finite if and only if there exists a polynomial p with $p(C) = 0$. As usual, if $p(x) = a_0 + a_1x + \cdots + a_nx^n$, then $p(C) = a_0I + a_1C + \cdots + a_nC^n$
(ii) Construct an example of a separable Hilbert space H and a compact linear map $C : H \to H$ whose spectrum is a finite set, but there is no polynomial p with $p(C) = 0$.